# Payment Model

Software Requirements / Logical Data Model

## 1. Purpose

Represents a hostel-related fee/payment record and its verification status.

## 2. Responsibilities

• Record the amount and context of a fee transaction or payment entry.

• Associate payment records with the relevant student and fee period/type where applicable.

• Support staff verification and fee-status reporting.

## 3. Proposed Logical Attributes

The following is a documentation proposal, not a claim that every field is already implemented. Confirm exact field names, types, constraints and cardinalities against the project ERD/database schema.

| Attribute | Meaning / Requirement |
|---|---|
| **payment_id** | Unique payment record identifier. |
| **student_id** | Reference to the student responsible for the fee. |
| **amount** | Payment amount, with currency/precision defined by policy. |
| **payment_date** | Date the payment was made or recorded. |
| **payment_method_or_reference** | Optional method/reference, if collected. |
| **status** | Pending, verified, rejected or other defined state. |
| **verified_by / verified_at** | Verification metadata, if maintained. |

## 4. Relationships

• A payment belongs to or is associated with a student.

• A payment may be associated with an allotment, fee period or fee type if the schema supports it.

• Payment records feed fee-status checks and administrative reports.

## 5. Validation and Business Rules

• Amounts must be valid non-negative monetary values; actual payment rules should specify whether zero is allowed.

• A payment should not be marked verified without the required verification process.

• The system should prevent duplicate recording of the same transaction when a reliable reference is available.

• Financial records should be protected and changes should be auditable.

## 6. Traceability / Notes

The initial scope includes maintaining hostel fee records and verifying payments; an online payment gateway is listed as outside the initial scope.

Status: Draft for review. Align this document with the approved SRS, ERD, class diagram, and institutional policies before treating it as final.